MIL-C-48490B(AR)
30 June 1981
SUPERSEDING
MIL-C-48490A(AR)
8 March 1976

MILITARY SPECIFICATION

CONTROL, COMMANDER'S SIGHT AND RECEIVER-TRANSMITTER, LASER RANGING: 11743121

This specification is approved for use by the US Army Armament Research and Development Command (ARRADCOM) and is available for use by all Departments and Agencies of the Department of Defense.

WARNING - EYE HAZARD

EXTREME CAUTION MUST BE OBSERVED IN OPERATING AND TESTING THE RECEIVER-TRANSMITTER. IF THE LASER LIGHT BEAM ENTERS THE HUMAN EYE, THE RETINA OF THE EYE CAN BE BURNED. THE HIGH ENERGY DENSITY OF THE BEAM CAN ALSO PRODUCE SUCH DAMAGE AT EXTREME RANGE AND BY REFLECTION FROM ANY SHINY SURFACE SUCH AS GLASS, METAL AND PAINT. IN VIEW OF THE POSSIBILITY OF PRODUCING SERIOUS EYE DAMAGE, THE RECEIVER-TRANSMITTER SHOULD NOT BE OPERATED UNLESS THE TARGET AREA HAS BEEN EVACUATED.

---

1. SCOPE

1.1 Scope. This specification establishes the requirements and quality assurance provisions for the Control, Commander's Sight and Receiver-Transmitter, Laser Ranging: 11743121 which is a unit of the Laser Rangefinder, 11743101.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

FSC: 1240

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:29Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

FEDERAL

BB-N-411	Nitrogen, Technical

MILITARY

MIL-O-13830	Optical Components for Fire Control Instruments: General Specification Governing the Manufacture, Assembly and Inspection of
MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-I-45607	Inspection Equipment: Acquisition, Maintenance and Disposition of
MIL-STD-45662	Calibration System Requirements

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-810	Environmental Test Methods

DRAWINGS

ARRADCOM

11743121	Control, Commander's Sight and Receiver-Transmitter, Laser Ranging

Inspection Equipment

11750222	Test Set - Laser System Test
11750232	Cabinet, Laser - FAT
11821748	Test Set - Immersion Leak, Purge and Fill

Packaging Data Sheet

P11743121	Packaging of Control, Commander's Sight and Receiver-Transmitter, Laser Ranging: 11743121

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:29Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Fabrication. The Control, Commander's Sight and Receiver-Transmitter, Laser Ranging, herein referred to as the assembly, shall be manufactured in accordance with Drawing 11743121 and drawings pertaining thereto and, when assembled, shall meet the requirements of this specification. (See 4.5.1)

3.1.1 Function. The assembly shall provide the following functions:

	Designation
a.	Laser system
b.	Sighting system
c.	Sight system and laser beam alignment
d.	Initial illumination dimmer control
e.	Lamp test control

3.1.2 General specifications. The following provisions of MIL-F-13926 apply: (See 4.5.1)

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishes
d. Part identification and marking
e. Workmanship

3.1.3 Ambient conditions. Standard ambient conditions shall be as follows:

a.	Temperature	$73^{\circ} \pm 18^{\circ}F$
b.	Relative humidity	50 percent ± 30 percent
c.	Atmospheric pressure	28.5 + 2.0 −3.0 in. Hg.

3.2 First article. When specified (see 6.2), the contractor shall furnish sample units for first article inspection and approval (see 4.4 and 6.2).

3.3 Performance. Unless otherwise specified, the assembly shall meet the performance requirements specified herein under standard ambient conditions of 3.1.3.

3.3.1 Laser system.

3.3.1.1 Continuity. With no power or signals applied, continuity shall exist between pins of connectors as specified in table II. (See 4.6.2.1)

3.3.1.2 Display readouts and lamp illumination. With signals applied to pins of J3 as listed in tables III and IV, display readouts and lamp illumination shall be as specified. (See 4.6.2.2)

3.3.1.3 Thermistor resistance. The thermistor resistance between J2-u and J2-v shall be 100 ± 60 kilohms. (See 4.6.2.3)

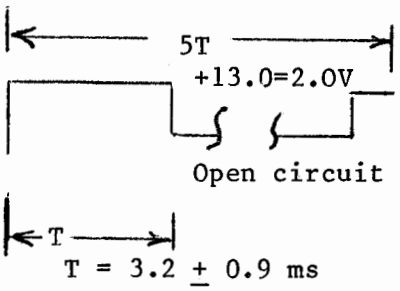
Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:29Z
Check the source to verify that this is the current version before use.

TABLE I. Loads, power and signals.

Item	Conditions	Characteristics	Connections
1.0	Loads		Unless otherwise specified, connected between the following pins:
1.1	Impedance	160 ohms ± 5 percent in series with 62 picofarads (pF) ± 10 percent	J2-j and J2-k
1.2	Impedance	160 ohms ± 5 percent in series with 62 pF ± 10 percent	J2-m and J2-n
2.0	Power sources		
2.1	400 V	400 ± 20 volts direct current (Vdc) Ripple: 100 millivolts (mV) max peak-to-peak (p-p)	J2-g (+) and J2-V (-)
2.2	29 V	29.0 ± 1.0 Vdc Ripple: 20 mV p-p max	J2-W (+) and J2-V (-)
2.3	24 V Q-switch	24.0 ± 2.0 Vdc Ripple: 200 mV p-p max	J2-S (+) and J2-V (-)
2.4	24 V	24.0 ± 2.0 Vdc Ripple: 200 mV p-p max	J2-L (+) and J2-E (-)
2.5	15 V	15.0 ± 0.7 Vdc Ripple: 25 mV p-p max	J2-T (+) and J2-V (-)
2.6	15 V Computer	15.0 ± 0.7 Vdc Ripple: 25 mV p-p max	J2-N (+) and J2-E (-)
2.7	14 V	14.0 ± 0.5 Vdc Ripple: 25 mV p-p max	J3-L (+) and J3-M (-)
2.8	5 V	5.0 ± 0.2 Vdc Ripple: 25 mV p-p max	J2-U (+) and J2-V (-) J3-K (+) and J3-M (-)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:29Z
Check the source to verify that this is the current version before use.

TABLE I. Loads, power and signals. - Continued

Item	Conditions	Characteristics	Connections
2.9	-5 V	-5.5 ± 0.2 Vdc Ripple: 25 mV p-p max	J2-X (-) and J2-V (+)
2.10	-15 V Computer	-15.0 ± 0.7 Vdc Ripple: 25 mV p-p max	J2-M (-) and J2-E (+)
2.11	-1600 V	-2180 ± 50 Vdc Ripple: 100 mV p-p max Source impedance = 480 kilohms ± 5 percent	J2-s (-) and J2-p (+)
2.12	115 V, 400 Hz	115.0 ± 5.0 volts root mean square (Vrms) 400 ± 4 Hz	J2-C and J2-D
2.13	AC Ref. 1	5.000 ± 0.002 Vrms 400 ± 4 Hz, 180° ± 6° out-of-phase with 115V, 400 Hz (item 2.12)	J2-B and J2-E
2.14	AC Ref. 2	5.000 ± 0.002 Vrms 400 ± 4 Hz, 180° ± 2.25° out-of-phase with AC Ref. 1	J2-A and J2-E
2.15	Gate waveform		Applied as specified herein
3.0	Signals sources		
3.1	Digital type-A	Logical one: 4.0=1.0 Vdc (source impedance: 2k ohms ± 5%) Logical zero: 0.2 ± 0.2 Vdc	Applied as specified herein

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:29Z
Check the source to verify that this is the current version before use.

TABLE I. Loads, power and signals. - Continued

Item	Conditions	Characteristics	Connections
3.2	Digital type-B	Logical one: open Logical zero: 0.2 ± 0.2 Vdc	Applied as specified herein
3.3	Analog 0 to 36 V	0 to 36 Vrms, 400 ± 4 Hz, in-phase or out-of-phase with 115V, 400 Hz source of item 2.12	J2-J and J2-K

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:29Z
Check the source to verify that this is the current version before use.

TABLE II. Continuity.

Switch	Switch Position	Continuity Between Pins J3-M and J3 Pins 1/									Continuity Between 1/			Continuity Between Pin J3-P and J3 Pins 1/		
		Y	Z	a	V	X	W	AA	U	s	J3-EE and J3-FF	J1-D and J2-FF	J2-GG and J2-HH	R	S	T
1	Normal	O	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
	Depressed	S	NA													
2	Normal	NA	O													
	Depressed		S	NA												
LAST	Normal		NA	O												
	Depressed			S	NA											
FEED	Normal			NA	O											
	Depressed				S	NA										
BATTLE RANGE	Normal				NA	O										
	Depressed					S	NA	NA								

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:29Z
Check the source to verify that this is the current version before use.

TABLE II. Continuity. - Continued

Switch	Switch Position	Continuity Between Pins J3-M and J3 Pins 1/									Continuity Between 1/			Continuity Between Pin J3-P and J3 Pins 1/		
		Y	Z	a	V	X	W	AA	U	s	J3-EE and J3-FF	J1-D and J2-FF	J2-GG and J2-HH	R	S	T
RANGE	Normal					NA	S	O								
	Depressed						O	S	NA							
RESET	Normal						NA	NA	O							
	Depressed								S					NA	NA	NA
MODE	TEST								NA					S	R	R
	ON								NA					R	S	R
	AUTO	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	R	R	S
Circuit Continuity		NA	NA	NA	NA	NA	NA	NA	NA	S	S	S	S	NA	NA	NA

1/ Continuity condition legend: O = open circuit - resistance 1 megohm. S = short circuit - resistance 1.0 ohm. R = 9.4 ± 2.0 kilohms. NA = not applicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:29Z
Check the source to verify that this is the current version before use.

TABLE III. Display readout.

Item	Input Conditions at Pins of J3 1/											Display Readouts 2/				
	A	B	C	D	q	e	J	H	G	F	E	1000's	100's	10's	1's	Returns
1	1	0	0	0	0	1	NC	NC	NC	NC	+	NI	NI	NI	NI	8
2	1	0	0	0	0	1	NC	NC	NC	$\overline{+}$	NC	NI	NI	NI	8	NI
3	1	0	0	0	0	1	NC	NC	+	NC	NC	NI	NI	8	NI	NI
4	1	0	0	0	0	1	NC	+	NC	NC	NC	NI	8	NI	NI	NI

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:29Z
Check the source to verify that this is the current version before use.

TABLE III. Display readout. - Continued

Item	Input Conditions at Pins of J3 1/											Display Readouts 2/				
	A	B	C	D	q	e	J	H	G	F	E	1000's	100's	10's	1's	Returns
5	1	0	0	0	0	1	+	NC	NC	NC	NC	8	NI	NI	NI	NI
6	1	0	0	0	1	1	+	NC	NC	NC	NC	NI	NI	NI	NI	NI
7	0	0	0	0	0	0	+	NC	NC	NC	NC	NI	NI	NI	NI	NI
8	0	1	1	0	0	1	+	NC	NC	NC	NC	6	NI	NI	NI	NI
9	0	0	1	1	0	1	+	NC	NC	NC	NC	3	NI	NI	NI	NI

1/ Input conditions legend: 1 = logical one digital type-A signal. 0 = logical zero digital type-A signal. + = item 2.15, table I. NC = no connection.

2/ Display readouts legend: NI = not illuminated.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:29Z
Check the source to verify that this is the current version before use.

TABLE IV. Lamp illumination.

Item	Conditions at Pins of J3 1/											Lamps Illuminated 2/
	m	p	b	c	d	h	g	i	j	k	n	
1	X											Select
2		X										MALF
3			X									Test
4				X								On
5					X							Auto
6						X						Battle Range

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:29Z
Check the source to verify that this is the current version before use.

TABLE IV. Lamp illumination. - Continued

Item	Conditions at Pins of J3 1/											Lamps Illuminated 2/
	m	p	b	c	d	h	g	i	j	k	n	
7							X					Range
8								X				First
9									X			Second
10										X		Last
11											X	Go

1/ X denotes lamp illumination when ground is applied to designated pin.
2/ The RESET and FEED lamps shall be continuously illuminated.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:29Z
Check the source to verify that this is the current version before use.

3.3.1.4 Lasing pulse timing. During laser firing, the time elapsed between the start of SCR trigger and the start of A-trigger, measured at J2-P and J2-n shall be as specified in figure 1. (See 4.6.2.4)

3.3.1.5 A-trigger waveforms. During laser firing, the A-trigger waveforms, measured at J2-m and J2-n shall be as specified in figure 2. (See 4.6.2.5)

3.3.1.6 Malfunction 3 signal. The voltage measured at J2-f shall be as follows: (See 4.6.2.6)

a. 4.2 ± 1.2 V after laser firing.

b. 0.2 ± 0.2 V with the voltage at J2-S removed and the logical zero type-A signal (item 3.1 of table I) momentarily applied to J2-R.

c. 4.2 ± 1.2 V with the logical zero type-A signal (item 3.1 of table I) momentarily applied to J2-e.

3.3.1.7 Range and video signal. With the digital type-A input signal (item 3.1 of table I) applied as specified in figure 3, the range and video signal outputs shall be as specified in figure 4. The signals may recur at a period of 1.0 ± 0.1 microseconds (ms). (See 4.6.2.7)

3.3.1.8 Reset signal. After laser firing, the reset signal as specified in figure 5 shall appear at J2-c and J2-d. (See 4.6.2.8)

3.3.1.9 AGC inhibit. With the input to the receiver aperture covered and the digital type-A input signal (item 3.1 of table I) applied as specified in figure 6, the AGC inhibit output between J2-b and J2-V (rtn) shall be 29.0 ± 4.0 V. With the type-A input signal removed, the output shall be 0.0 ± 1.0 V. (See 4.6.2.9)

3.3.1.10 Ranging. Range conditions of the assembly shall be as follows: (See 4.6.2.10)

a. The minimum range shall be 200 meters.
b. The maximum range shall be 5000 meters.
c. The range accuracy shall be plus or minus 10 meters.
d. The range resolution shall be 20 meters.

3.3.1.11 Minimum range inhibit. With the optical input signal to the receiver aperture attenuated 40 ± 1 decibels (dB) for Target No. 3A or 36 ± 1 dB for Target No. 3B and the logical zero type-B signal (item 3.2 of table I) applied to J2-w, the second video pulse signal as specified in figure 7 shall not be present with the laser fired at Target No. 3A or 3B (item 8 of table X) and the RANGE readouts shall display 9990, 9995 or 000 and the RETURNS readout shall display 0. (See 4.6.2.11)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:29Z
Check the source to verify that this is the current version before use.

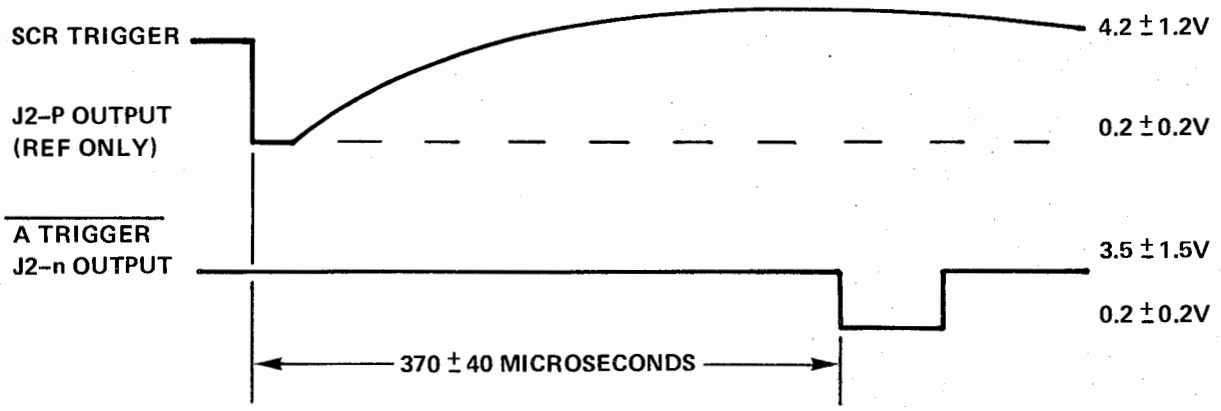


FIGURE 1. SCR trigger and A-trigger timing waveforms.

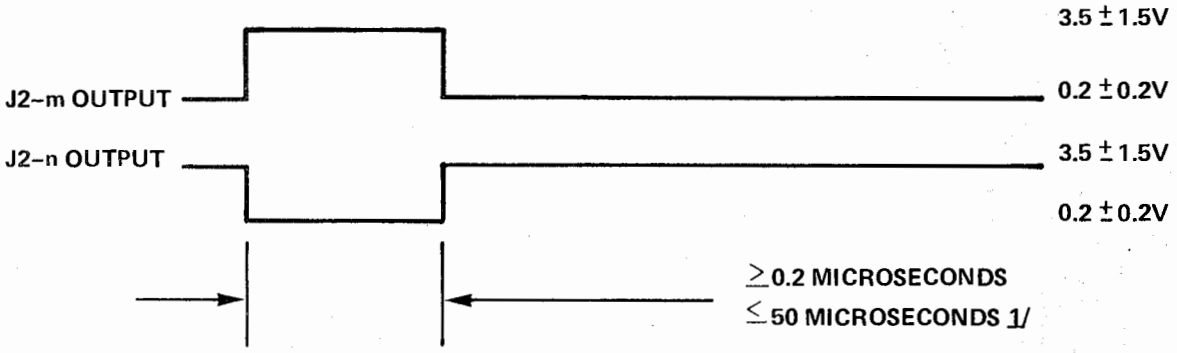


FIGURE 2. A-trigger waveforms.

1/ In figure 2 and subsequent figures where applicable, the pulse width is specified at 50 percent of peak pulse amplitude.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:29Z
Check the source to verify that this is the current version before use.

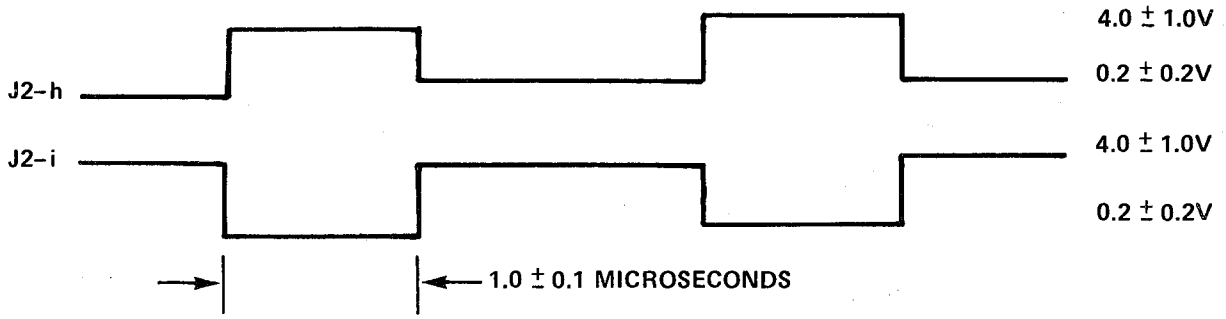


FIGURE 3. Test range and video output waveforms.

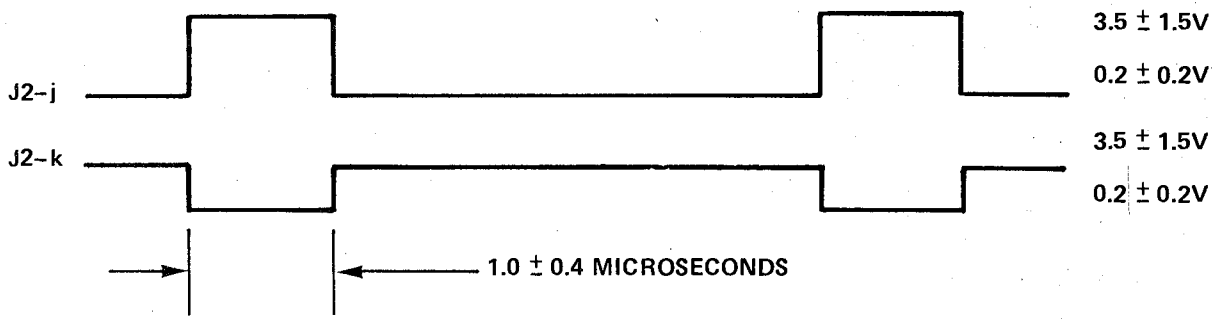


FIGURE 4. Test range and video output waveforms.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:29Z
Check the source to verify that this is the current version before use.

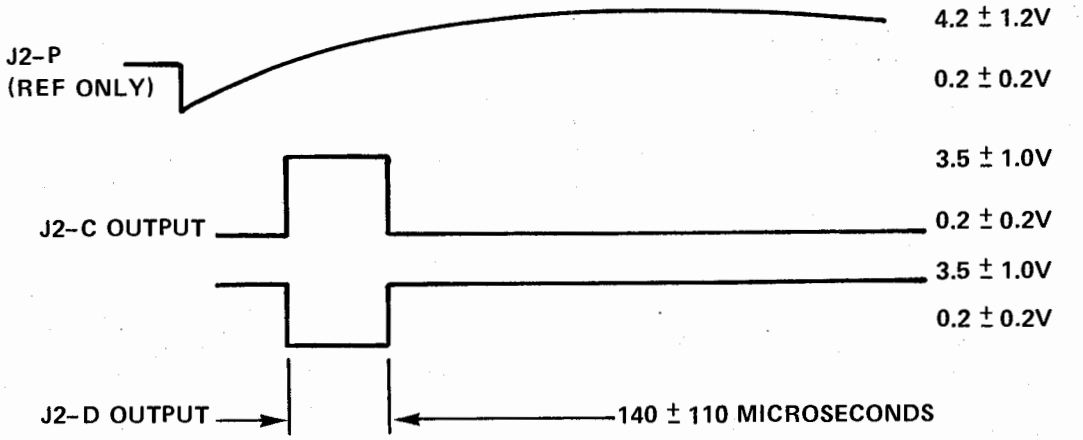


FIGURE 5. Reset signal waveforms.

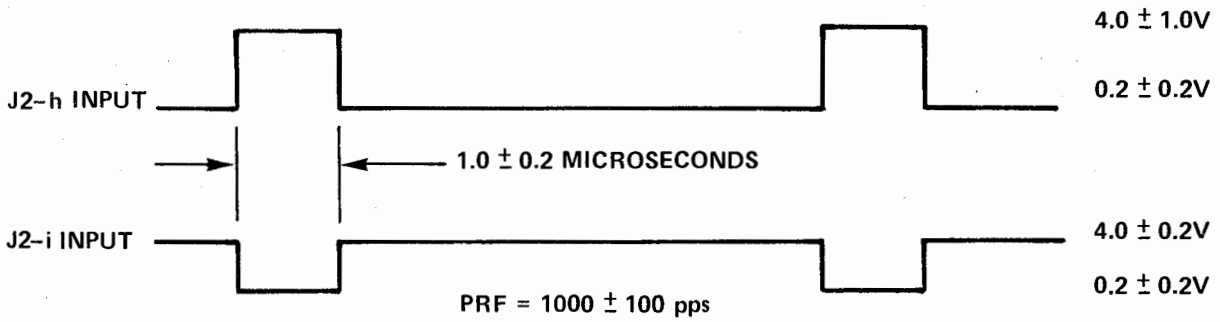


FIGURE 6. AGC inhibit input signal waveforms.

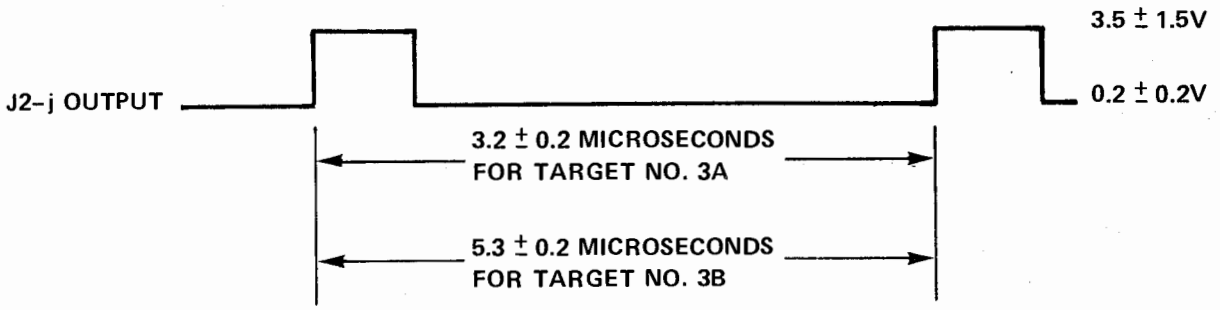


FIGURE 7. Ranging video signal waveform.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:29Z
Check the source to verify that this is the current version before use.

3.3.1.12 Events counter. The events counter shall record one event for each activation of the Q-switch motor or flashtube. (See 4.6.2.12)

3.3.1.13 A-trigger threshold and PFN low charge. With the energy output of the assembly set to 15 ± millijoules and after laser firing, the voltage of J2-f shall be 3.5 ± 1.0 Vdc. With 10.0 ± 1.0 Vdc applied to J2-x after laser firing, the voltage at J2-f shall be 0.2 ± 0.2 Vdc. (See 4.6.2.13)

3.3.1.14 Laser energy. With the energy output of the assembly set to 50 ± 5 millijoules, the PFN voltage sense current shall be less than that specified in figure 8 for the corresponding thermistor resistance and assembly temperature. (See 4.6.2.14)

3.3.1.15 Laser peak power. Laser peak power output of the assembly shall be not less than 1.0 megawatt. (See 4.6.2.15)

3.3.2 Sight system. The boresight position referred to in the following requirements is the condition in which the line of sight through the reticle is perpendicular to the plane through the axis of rotation and the ballistic drive linkage bearing centerline.

3.3.2.1 Boresight controls - elevation. (See 4.6.2.16)

3.3.2.1.1 Boresight knob travel. The elevation boresight knob travel shall be not less than ± 5.0 mils from the boresight position defined in 3.3.2. Clockwise rotation of the elevation boresight knob shall move the reticle upward.

3.3.2.1.2 Boresight knob accuracy. With rotation of the boresight knob clockwise and counterclockwise there shall be a corresponding movement of the reticle within ± 0.2 mil throughout the +5.0 mils travel requirement.

3.3.2.1.3 Backlash. Backlash in the elevation boresight knob shall be not greater than 0.1 mil.

3.3.2.1.4 Reticle shift. Azimuth shift of the reticle shall be not greater than 0.1 mil with rotation of the boresight knob ± 5.0 mils from the boresight position.

3.3.2.2 Boresight controls - azimuth. With the 5.000 ± 0.002 Vrms AC Ref 1 and AC Ref 2 (items 2.13 and 2.14 of table I) applied as specified, the azimuth boresight controls shall meet requirements of 3.3.2.2.1 through 3.3.2.2.3. (See 4.6.2.17)

3.3.2.2.1 Null position. With the assembly set at the boresight condition defined in 3.3.2, the azimuth boresight knob shall be capable of adjustment to achieve a null output at J2-AA of not greater than 10 mV rms.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:29Z
Check the source to verify that this is the current version before use.

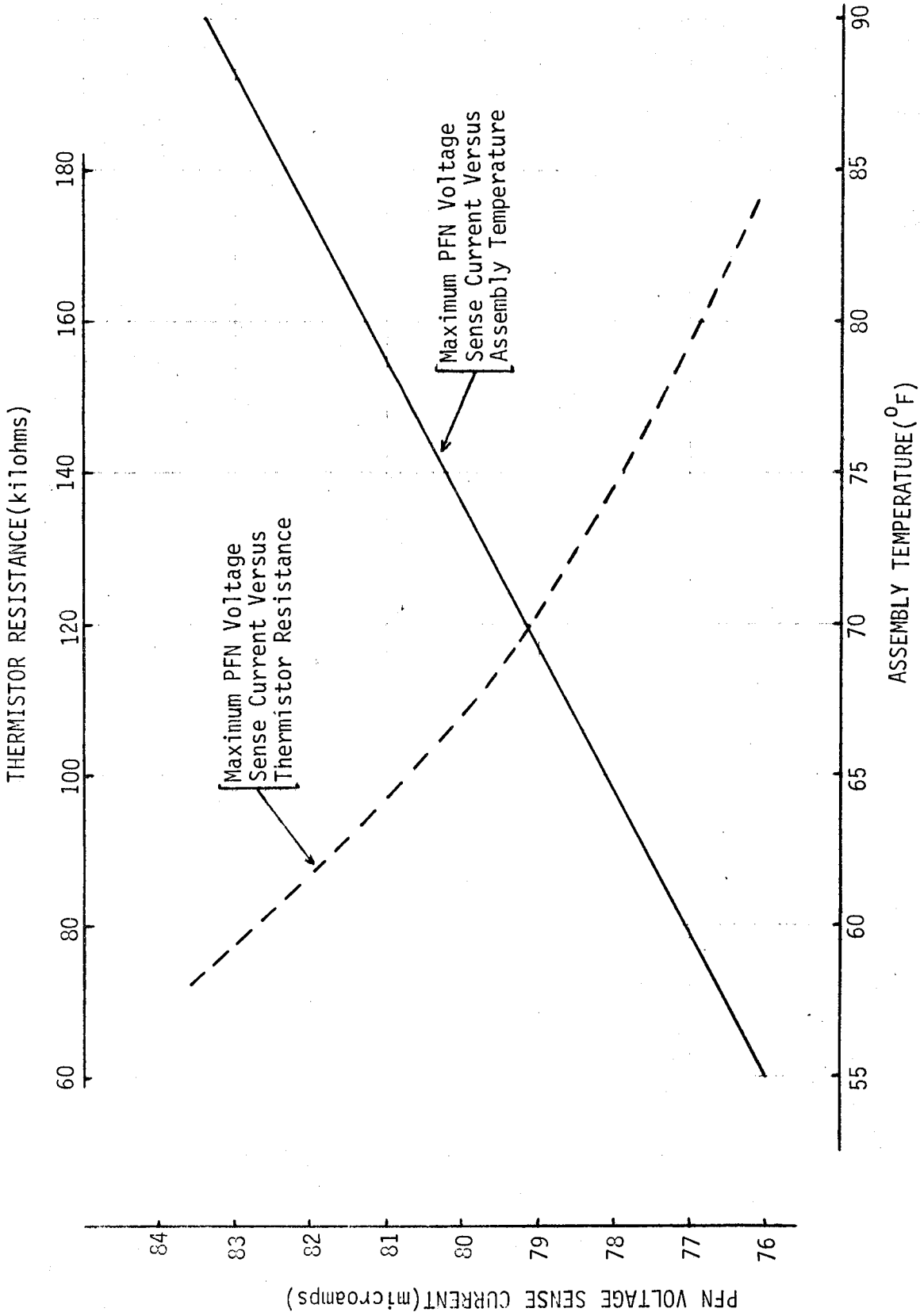


FIGURE 8. PFN sense current characteristics.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:29Z
Check the source to verify that this is the current version before use.

3.3.2.2.2 Boresight knob travel. With the mirror positioned to achieve the boresight position defined in 3.3.2 and the azimuth knob adjusted for the null condition defined in 3.3.2.2.1, the boresight knob shall be further adjustable clockwise and counterclockwise to obtain a feedback of not less than 295 mV rms (5.0 mils) measured at J2-AA. With clockwise rotation of the boresight knob, the feedback voltage shall be in phase with AC Ref 1 and with counterclockwise rotation, the feedback voltage shall be out of phase with AC Ref 1.

3.3.2.2.3 Boresight knob accuracy. Graduations on the boresight knob shall be accurate to produce feedback voltages within ± 9.0 mV rms throughout the boresight knob travel requirement of 3.3.2.2.2 at a scale factor of 58.9 mV rms per mil graduation of the knob.

3.3.2.3 Mirror drive controls. With the 5.000 ± 0.002 Vrms AC Ref 1 and AC Ref 2 (items 2.13 and 2.14 of table I) applied as specified, the mirror drive controls shall meet the requirements of 3.3.2.3.1 through 3.3.2.3.6. (See 4.6.2.18)

3.3.2.3.1 Mirror (resolver) accuracy. From the null position of 3.3.2.2.1 the resolver feedback voltage shall be as specified in table V for each target position defined.

TABLE V. Laser feedback.

Item	Target position in mils	Laser feedback (mVrms) at pins J2-AA and J2-E (rtn)
1	0	0 ± 10
2	12 Right	707 ± 10
3	24 Right	1414 ± 10 In Phase with AC Ref 1
4	36 Right	2121 ± 18
5	0	0 ± 10
6	12 Left	707 ± 10
7	24 Left	1414 ± 10 Out-of-Phase with AC Ref 1
8	36 Left	2121 ± 18
9	0	0 ± 10

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:29Z
Check the source to verify that this is the current version before use.

3.3.2.3.2 Reticle shift (elevation). With a target line-of-sight deviation of 36 mils to the right and left of the boresight position defined in 3.3.2 the reticle shall deviate not greater than 0.15 mil in elevation during each 36 mils of travel from a line parallel to the assembly axis of rotation.

3.3.2.3.3 Mirror slew rate. The mirror shall be capable of rotating its full 36 mils travel, equivalent to 72 mils target line-of-sight deviation, within 2.2 seconds. At temperature extremes of -25°F and +125°F, rotation through the full 36 mils travel shall be within 3 seconds.

3.3.2.3.4 Tachometer outputs. With no voltage applied to the mirror drive motor, tachometer output at J2-H (J2-G rtn) shall be not greater than 40.0 mV rms. With the motor drive input set to 36.0 ± 1.0 Vrms at 400 Hz, the tachometer output at J2-H (J2-G rtn) shall be not less than 2.5 Vrms and shall be in phase opposition to the motor drive input (J2-J, J2-K (ref)).

3.3.2.3.5 Mirror - servo system stability. With power and signals applied to the assembly, the mirror oscillation shall be less than ± 0.05 mil peak-to-peak.

3.3.2.3.6 Mirror - servo system overshoot. With power and signals applied to the assembly, the mirror shall drive to the positions defined in table I applied to the assembly and the mirror oscillation shall be less than ± 0.05 mil peak-to-peak.

3.3.2.4 Resolution. The resolution on the optical axis shall be not greater than 7 arcseconds at 12X and shall be not greater than 10 arcseconds at 6X. The resolution at 0.7 field shall be not greater than 15 arcseconds at 12X and shall be not greater than 30 arcseconds at 6X. (See 4.6.2.19)

3.3.2.5 Parallax. With a target viewed at a range of 1000 ± 100 meters, parallax at the center of field shall be not greater than 0.1 mil. (See 4.6.2.20)

3.3.2.6 Cleanliness. The assembly shall meet the assembled optical system cleanliness requirements specified in MIL-O-13830. (See 4.6.2.21)

3.3.2.7 Image tilt. The sighting system shall rotate the image of an object not greater than two degrees. (See 4.6.2.22)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:29Z
Check the source to verify that this is the current version before use.

3.3.2.8 Parallelism of reticle and image. The image of a line parallel to the assembly axis of rotation shall be parallel to the center horizontal reticle line within one degree. (See 4.6.2.23)

3.3.2.9 Eyepiece focus. Focus of the eyepiece shall be adjustable on a diopter scale from plus 2.8 diopters to minus 2.8 diopters. (See 4.6.2.24)

3.3.2.10 Reticle illumination. With the reticle light control set for normal illumination, the reticle pattern shall be clearly defined in a darkened area. Clockwise rotation of the reticle light control shall increase the intensity of the reticle illumination. (See 4.6.2.25)

3.3.2.11 Operability. (See 4.6.2.26)

3.3.2.11.1 Diopter adjustment torque. The torque required to adjust the diopter setting shall be within the limits of 3.0 and 18.0 inch-pounds under the conditions of 3.1.3 and shall be not greater than 20.0 inch-pounds at -25°F and 125°F.

3.3.2.11.2 Boresight knob torque. The torque required to rotate the boresight knobs shall be within the limits of 0.5 and 8.0 inch-pounds under the conditions of 3.1.3 and shall be not greater than 12.0 inch-pounds at -25°F and 125°F.

3.3.2.11.3 Boresight knob disengagement. The force required to disengage the boresight knobs from their normal position shall be within the limits of 5.0 and 12.0 pounds under the conditions of 3.1.3 and shall be not greater than 18.0 pounds at -25°F and 125°F.

3.3.2.11.4 Magnification knob torque. The torque required to change magnification shall be within the limits of 6.0 and 12.0 inch-pounds under the conditions of 3.1.3 and shall be not greater than 18.0 inch-pounds at -25°F and 125°F.

3.3.2.12 Boresight retention. An established boresight position shall change not greater than ±0.25 mil in azimuth and ±0.35 mil in elevation under the conditions of 3.1.3. Under storage temperature conditons (-65°F to 160°F) the change in boresight shall be not greater than ±0.5 mil in azimuth and ±0.7 mil in elevation. (See 4.6.2.27)

3.3.3 Sight system and laser beam alignment. (See 4.6.2.28)

3.3.3.1 Iniital alignment. The transmitted laser beam shall be parallel to the reticle centerline within 0.1 mil.

3.3.3.2 Alignment retention. The assembly shall retain the laser beam alignment established in 3.3.3.1 within 0.25 mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:29Z
Check the source to verify that this is the current version before use.

3.3.4 Indicator illumination dimmer control. With power sources of table I applied and with the dimmer control knob turned clockwise toward but not into the TEST position, the FEED and RESET indicator lamps shall change from dim to bright. With the knob turned in the counterclockwise direction, the indicator lamps shall change from bright to dim. (See 4.6.2.29)

3.3.5 Lamp test control. With the power source (item 2.15, table I) applied to pin J3-J, with the return connected to J3-M, with a logical zero digital type-A signal (item 3.1, table I) applied to J3-A and J3-B with a logical one digital type-A signal applied to J3-C and J3-D and with the dimmer control rotated fully clockwise to the TEST position, the digit 8 shall appear in the thousands readout and the voltage between pins J3-R and J3-M (rtn) shall be 0.2 $\pm$ 0.2V. With the dimmer control knob turned counterclockwise out of the TEST position, the readout shall display the digit 3 and the output voltage between pins J3-r and J3-M (rtn) shall be 4.2 $\pm$ 1.2V. (See 4.6.2.30)

3.4 Sealing. The optical and transmitter compartments shall be purged with dry nitrogen in accordance with BB-N-411, type I, grade B, class 1 with a dew point at least as low as -25°F, at a pressure of 12 $\pm$ 1 psig for a 30 minute period and with a final pressure of 4.5 $\pm$ 0.5 psig. The electronic compartments shall be purged at 1 psig for a five minute period with a final pressure of 0.0 +0.2 - 0.0 psig. With an initial internal pressure of 4.5 $\pm$ 0.5 psig in each compartment, the pressure drop of each compartment shall be not greater than 0.2 psig in a 2-hour period. (See 4.6.3)

3.5 Environmental.

3.5.1 Shock. The assembly shall be capable of operating as specified herein after exposure under the conditions of 3.1.3 to three half sine wave shock pulses of 40 $\pm$ 4 gravity unit (g) for a duration of 18 $\pm$ 3 milliseconds (msec) applied in each direction along three mutually perpendicular axes. In addition, the assembly shall be capable of withstanding three half sine wave shock pulses of 100.0 $\pm$ 10.0 g for a duration of 1.5 $\pm$ 0.2 msec applied in each direction along three mutually perpendicular axes. (See 4.6.4.1)

3.5.2 Vibration. The assembly shall be capable of operating as specified herein after exposure under the conditions of 3.1.3 to the vibration profile of figure 9. Duration of exposure shall be not less than 80 minutes in each of three mutually perpendicular axes. (See 4.6.4.2)

3.5.3 Operating temperature. The assembly shall be capable of operating as specified herein over the operating temperature range of -25°F to +125°F. (See 4.6.4.3, 4.6.4.4 and 4.6.4.5)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:29Z
Check the source to verify that this is the current version before use.

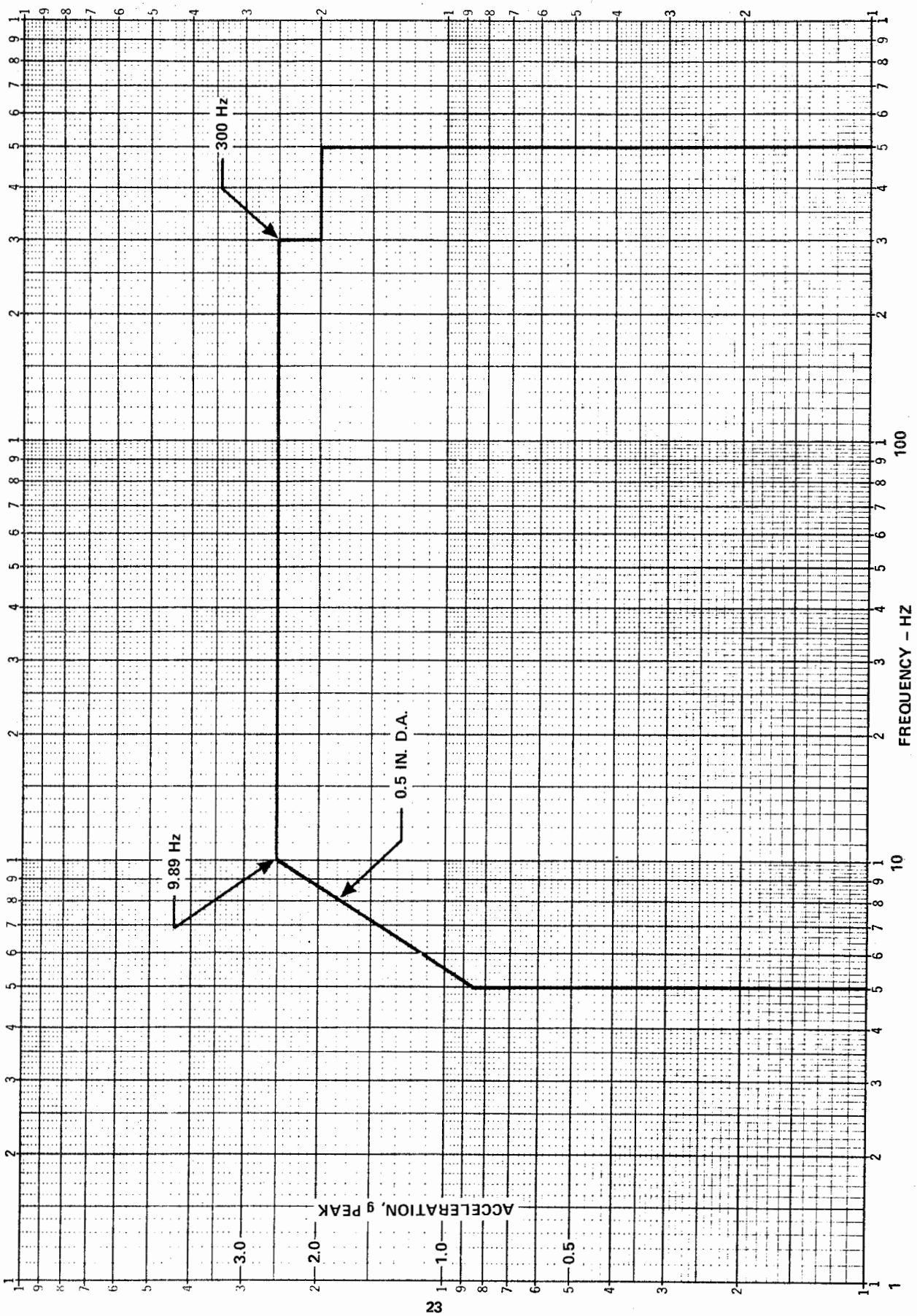


FIGURE 9. Vibration profile.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:29Z
Check the source to verify that this is the current version before use.

3.5.4 Storage temperature. The assembly shall be capable of operating as specified herein after exposure to storage temperatures ranging from -65°F to 160°F. (See 4.6.4.3, 4.6.4.4 and 4.6.4.5)

3.5.5 Humidity. The assembly shall operate as specified herein at up to 100 percent relative humidity. (See 4.6.4.6)

3.5.6 Altitude. The assembly shall operate as specified herein under the following conditions:

a. during exposure to an atmospheric pressure equivalent to 10000 feet above sea level.

b. after exposure to an atmospheric pressure equivalent to 40000 feet above sea level for not less than 1 hour. (See 4.6.4.7).

3.5.7 Dust. The assembly shall operate as specified herein after exposure to moving dust for not less than 12 hours at velocities up to 1750 feet per minute. (See 4.6.4.8)

3.5.8 Salt fog. The assembly shall operate as specified herein after exposure to a 5 percent sodium chloride atomized spray for not less than 48 hours. (See 4.6.4.9)

4. QUALITY ASSURANCE PROVISIONS

SAFETY PRECAUTIONS

These limitations are recommended for the protection of personnel against potential hazards from backscattered laser radiation:

1. Conduct no test in rain, snow or hail.

2. Equip all personnel using the rangefinder and all personnel in the vicinity with laser safety glasses of at least density 4 at 6943 angstroms when testing under these conditions:

   a. Morning twilight
   b. Evening twilight
   c. Daytime heavy fog where visibility is less than 200 meters.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:29Z
Check the source to verify that this is the current version before use.

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classified of inspections. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.4).
2. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the quality assurance provisions of MIL-F-13926 and the conditions of 3.1.3.

4.4 First article. The requirement for first article approval and the designation of responsibility for first article inspection to either the Government or the contractor shall be specified in the contract.

4.4.1 Sample. The first article sample shall be three assemblies selected at random by the Government representative from the first fifteen production assemblies.

4.4.2 Inspection. The sample shall be subjected to all the inspections specified in table VI.

4.4.3 Failure. Failure of any assembly to meet any requirement shall be cause for refusal to grant first article approval. The Government reserves the right to terminate first article inspection upon any failure of any assembly to comply with any stated requirement.

4.4.4 Responsibility. The contractor, whether or not responsible, shall inspect the sample for conformance to all contractual requirements and shall submit a record of this inspection with the sample and certificates of conformance for materials. The Government reserves the right to witness inspections performed by the contractor.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:29Z
Check the source to verify that this is the current version before use.

TABLE VI. First article tests.

Item	Characteristic	Requirement	Test Procedure
1	Fabrication	3.1	Applicable drawings - Visual
2	General specifications	3.1.2	MIL-F-13926 - Visual
3	Performance	3.3	4.6.2
4	Sealing A and B	3.4	4.6.3
5	Shock	3.5.1	4.6.4.1
6	Vibration A	3.5.2	4.6.4.2
7	High temperature	3.5.3 and 3.5.4	4.6.4.3
8	Low temperature	3.5.3 and 3.5.4	4.6.4.4
9	Temperature cycling	3.5.3 and 3.5.4	4.6.4.5
10	Humidity	3.5.5	4.6.4.6
11	Altitude	3.5.6	4.6.4.7
12	Dust	3.5.7	4.6.4.8
13	Salt fog	3.5.8	4.6.4.9

4.5 Quality conformance inspection.

4.5.1 Procurement conditions. The following inspection plans shall apply where the assembly is procured for use as a:

Use	Plan
a. Logistics spare assembly apart from the next higher assembly	A and C
b. Component assembly of the next higher assembly	B and C

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:29Z
Check the source to verify that this is the current version before use.

4.5.2 Inspection plan A.

4.5.2.1 General sample and tests. One assembly, as a control sample, shall be selected at random by the Government representative from each 100 assemblies produced and shall be subjected to all the tests in table VIII.

4.5.2.2 Environmental sample and tests. Two assemblies, as a control sample, shall be selected at random by the Government representative from each 50 assemblies produced or from each month's production, whichever occurs first. All the tests in table IX shall be applied separately to each assembly in the sample.

4.5.2.3 Acceptance. Where any one assembly of either sample fails to meet any specified requirement, the lot shall be rejected. Rejected lots shall be subjected to the provisions of MIL-STD-105.

4.5.3 Inspection plan B.

4.5.3.1 Sample and tests. The sample shall be selected by the Government representative in accordance with the provisions of MIL-STD-105. All the tests in table VIII and only the test of item 307 in table IX shall be applied.

4.5.3.2 Acceptance. Acceptance and rejection shall be in accordance with MIL-STD-105.

4.5.4 Inspection plan C.

4.5.4.1 Sample and tests. Each assembly in every lot shall be subjected to all the tests in table VII and shall be examined visually for completeness, improper assembly and evidence of poor workmanship.

4.5.4.2 Acceptance. Where any one assembly fails to meet any specified requirement, the defective assembly shall be removed from the lot and resubmitted only after all defects have been corrected.

4.5.5 Inspection equipment. Unless otherwise specified in the contract, the contractor shall supply, maintain and calibrate inspection equipment in accordance with the provisions of MIL-I-45607 and MIL-STD-45662.

4.5.6 Packaging inspection. The sampling and inspection of the preservation - packaging, packing and container marking shall be in accordance with the provisions of packaging data sheet P11743121.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:29Z
Check the source to verify that this is the current version before use.

TABLE VII. Performance tests.

Item	Characteristic	Requirement	Test Procedure
101	Continuity	3.3.1.1	4.6.2.1
102	Display readouts and lamp illumination	3.3.1.2	4.6.2.2
103	Thermistor resistance	3.3.1.3	4.6.2.3
104	Lasing pulse timing	3.3.1.4	4.6.2.4
105	A-trigger waveforms	3.3.1.5	4.6.2.5
106	Malfunction 3 signal	3.3.1.6	4.6.2.6
107	Range and video signal	3.3.1.7	4.6.2.7
108	Reset signal	3.3.1.8	4.6.2.8
109	AGC inhibit	3.3.1.9	4.6.2.9
110	Ranging	3.3.1.10	4.6.2.10
111	Minimum range inhibit	3.3.1.11	4.6.2.11
112	Events counter	3.3.1.12	4.6.2.12
113	A-trigger threshold and PFN low charge	3.3.1.13	4.6.2.13
114	Laser energy	3.3.1.14	4.6.2.14
115	Laser peak power	3.3.1.15	4.6.2.15
116	Boresight controls - elevation	3.3.2.1	4.6.2.16
117	Boresight controls - azimuth	3.3.2.2	4.6.2.17
118	Mirror drive controls	3.3.2.3	4.6.2.18
119	Resolution	3.3.2.4	4.6.2.19
120	Parallax	3.3.2.5	4.6.2.20
121	Cleanliness	3.3.2.6	4.6.2.21
122	Image tilt	3.3.2.7	4.6.2.22
123	Parallelism of reticle and image	3.3.2.8	4.6.2.23
124	Eyepiece focus	3.3.2.9	4.6.2.24
125	Reticle illumination	3.3.2.10	4.6.2.25
126	Operability	3.3.2.11	4.6.2.26
127	Boresight retention	3.3.2.12	4.6.2.27
128	Sight system and laser beam alignment	3.3.3	4.6.2.28
129	Indicator illumination dimmer control	3.3.4	4.6.2.29
130	Lamp test control	3.3.5	4.6.2.30
131	Sealing A	3.4	4.6.3

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:29Z
Check the source to verify that this is the current version before use.

TABLE VIII. General tests.

Item	Characteristic	Requirement	Test Procedures
301	Fabrication	3.1	Applicable drawings - Visual
302	General specifications	3.1.2	MIL-F-13926 - Visual

TABLE IX. Environmental tests.

Item	Characteristic	Requirement	Test Procedure
303	Shock	3.5.1	4.6.4.1
304	Vibration B	3.5.2	4.6.4.2
305	High Temperature	3.5.3 and 3.5.4	4.6.4.3
306	Low Temperature	3.5.3 and 3.5.4	4.6.4.4
307	Temperature cycling	3.5.3 and 3.5.4	4.6.4.6
308	Sealing B	3.4	4.6.3

4.6 Methods of inspection.

4.6.1 Test equipment. The test equipment in table X shall be used where specified to perform the required test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:29Z
Check the source to verify that this is the current version before use.

TABLE X. Test equipment.

Item	Part Number	Nomenclature
1	11750222	Test Set - Laser System Test
2	11750232	Cabinent, Laser - FAT
3	11821748	Test Set - Immersion Leak, Purge and Fill
4	Target 1	12-inch minimum diameter circle, 15 percent minimum diffuse reflectivity and located at 215 ± 5 meters from window of receiver-transmitter. The exact location of the target within this tolerance shall be known within ±1 meter.
5	Target 2	23-inch minimum diameter circle, 15-percent minimum diffuse reflectivity and located at the actual distance of Target 3 minus 20 meters, from window of receiver-transmitter. The exact location of the target within this tolerance shall be known within ±3 meter.
6	Target 3A	4 feet square minimum, 15 percent minimum diffuse reflectivity, unless otherwise specified, and located at 490 ± 15 meters from window of receiver-transmitter. The exact target location within this tolerance shall be known within ±1 meter.
7	Target 3B	4 feet square minimum, 15 percent minimum diffuse reflectivity and located at 810 ± 15 meters from window of Receiver-Transmitter. The exact location of the target within this tolerance shall be known within ±3 meter.
8	Target 4A	4 feet square minimum, 80 percent minimum diffuse reflectivity and located at 490 ± 15 meters from window of Receiver-Transmitter. The exact location of the target within this tolerance shall be known within ±4 meter.
9	Target 4B	4 feet square minimum, 80 percent minimum diffuse reflectivity and located at 810 ± 15 meters from window of Receiver-Transmitter. The exact location of the target within this tolerance shall be known within ±3 meter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:29Z
Check the source to verify that this is the current version before use.

4.6.2 Performance tests.

4.6.2.1 Continuity. Use item 1 of table X to determine compliance with 3.3.1.1.

4.6.2.2 Display readouts and lamp illumination. Use item 1 of table X to determine compliance with 3.3.1.2.

4.6.2.3 Thermistor resistance. Use item 1 of table X to determine compliance with 3.3.1.3.

4.6.2.4 Lasing pulse timing. Use item 1 of table X to determine compliance with 3.3.1.4.

4.6.2.5 A-trigger waveforms. Use item 1 of table X to determine compliance with 3.3.1.5.

4.6.2.6 Malfunction 3 signal. Use item 1 of table X to determine compliance with 3.3.1.6.

4.6.2.7 Range and video signal. Use item 1 of table X to determine compliance with 3.3.1.7.

4.6.2.8 Reset signal. Use item 1 of table X to determine compliance with 3.3.1.8.

4.6.2.9 AGC inhibit. Use item 1 of table X to determine compliance with 3.3.1.9.

4.6.2.10 Ranging. Use items 1, 4, 5, 6, 7, 8 and 9 of table X to determine compliance with 3.3.1.10.

4.6.2.11 Minimum range inhibit. Use items 1, 6 and 7 of table X to determine compliance with 3.3.1.11.

4.6.2.12 Events counter. Use item 1 of table X to determine compliance with 3.3.1.12.

4.6.2.13 A-trigger threshold and PFN low charge. Use item 1 of table X to determine compliance with 3.3.1.13.

4.6.2.14 Laser energy. Use item 1 of table X to determine compliance with 3.3.1.14.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:29Z
Check the source to verify that this is the current version before use.

4.6.2.15 Laser peak power. Use item 1 of table X to determine compliance with 3.3.1.15.

4.6.2.16 Boresight controls - elevation. Use item 1 of table X to determine compliance with 3.3.2.1.

4.6.2.17 Boresigh controls - azimuth. Use item 1 of table X to determine compliance with 3.3.2.2.

4.6.2.18 Mirror drive controls. Use item 1 of table X to determine compliance with 3.3.2.3.

4.6.2.19 Resolution. Use item 1 of table X to determine compliance with 3.3.2.4.

4.6.2.20 Parallax. Use item 1 of table X to determine compliance with 3.3.2.5.

4.6.2.21 Cleanliness. The assembly shall be examined in accordance with the assembled optical system cleanliness quality assurance provisions of MIL-O-13830.

4.6.2.22 Image tilt. Use item 1 of table X to determine compliance with 3.3.2.7.

4.6.2.23 Parallelism of reticle and image. Use item 1 of table X to determine compliance with 3.3.2.8.

4.6.2.24 Eyepiece focus. Use item 1 of table X to determine compliance with 3.3.2.9.

4.6.2.25 Reticle illumination. Use item 1 of table X to determine compliance with 3.3.2.10.

4.6.2.26 Operability. Use item 1 of table X to determine compliance with 3.3.2.11.

4.6.2.27 Boresight retention. Use item 1 of table X to determine compliance with 3.3.2.12.

4.6.2.28 Sight system and laser beam alignment. Use item 1 of table X to determine compliance with 3.3.3.

4.6.2.29 Indicator illumination dimmer control. Use item 1 of table X to determine compliance with 3.3.4.

4.6.2.30 Lamp test control. Use item 1 of table X to determine compliance with 3.3.5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:29Z
Check the source to verify that this is the current version before use.

4.6.3 Sealing. Pressurize each compartment with dry nitrogen to 4.5 ± 0.5 psig and perform either Sealing A or Sealing B leakage test. After determination of compliance with the sealing requirement, purge the optical and transmitter compartments with dry nitrogen at 12 ± 1 psig for not less than 30 minutes, purge the compartment behind the commander's panel with dry nitrogen at 1 ± 0.1 psig for not less than 5 minutes, lower the pressure of the optical and transmitter compartments to 4.5 ± 0.5 psig and lower the pressure of the compartment behind the commander's panel to 0.0 ± 0.2 - 0.0 psig. Nitrogen shall be BB-N-411, type I, class 1, grade B with a dewpoint not greater than -25°F.

4.6.3.1 Sealing A. Use item 3 of table X and subject the assembly to the leakage (immersion) test specified in MIL-STD-810, method 512.1, procedure IV. Immersion time shall be not less than 5 minutes.

4.6.3.2 Sealing B. Use item 3 of table X, place the assembly in the following environment for not less than 2 hours and verify that the pressure drop is not greater than 0.2 psig.

a.	Temperature	73° ± 2.5°F
b.	Relative humidity	50 percent ± 5 percent
c.	Atmospheric pressure	28.5 + 2.0 -3.0 in. Hg

4.6.4 Environmental tests.

4.6.4.1 Shock.

4.6.4.1.1 Basic design. Use item 2 of table X and subject the assembly to the shock test specified in MIL-STD-810, method 516.2, procedure I, figure 516.2-2. Apply three half sine wave shock pulses in each direction along the three axes. Peak amplitude shall be 40 ± 4 g with a time duration of 18 ± 3 msec measured at the 10 percent amplitude points. At the conclusion of this test, subject the assembly to the performance tests of table VII.

4.6.4.1.2 High intensity. Use item 2 of table X and subject the assembly to the shock test specified in MIL-STD-810, method 516.2, procedure IV, figure 516.2-2. Apply three half sine wave shock pulses in each direction along the three axes. Peak amplitude shall be 100.0 ± 10.0 g with a time duration of 1.5 ± 0.2 msec measured at the 10 percent amplitude points. At the conclusion of this test, subject the assembly to the performance tests of table VII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:29Z
Check the source to verify that this is the current version before use.

4.6.4.2 Vibration.

4.6.4.2.1 Vibration A. Subject the assembly to the vibration test specified in MIL-STD-810, method 514.2 procedure VIII, except that the test level shall be the vibration curve shown in figure 9 and duration of exposure shall be not less than 80 minutes per axis. At the conclusion of this test, subject the assembly to the performance tests of table VII.

4.6.4.2.2 Vibration B. This test shall be conducted as in 4.6.4.2.1 except that the period of vibration shall be not less than 15 minutes in each axis.

4.6.4.3 High temperature. Subject the assembly to the high temperature test specified in MIL-STD-810, method 501.1, procedure I. The highest operating temperature shall be 140°F. The performance tests of table VII shall be applied.

4.6.4.4 Low temperature. Subject the assembly to the low temperature test specified in MIL-STD-810, method 502.1, procedure I. The storage temperature shall be -65°F. The lowest operating temperature shall be -25°F. The performance tests of table VII shall be applied.

4.6.4.5 Temperature cycling. With power and loads of table I applied, subject the assembly to the temperature profile specified in figure 10. At the conclusion of this test, subject the assembly to the performance tests of table VII.

4.6.4.6 Humidity. Place the assembly in a humidity chamber and subject it to the humidity test specified in MIL-STD-810, method 507.1, procedure IV. During steps 4, 6 and 8 of procedure IV, subject the assembly to the performance tests of 4.6.2. After exposure, subject the assembly to the performance tests of table VII.

4.6.4.7 Altitude. Place the assembly in an altitude chamber and subject it to the altitude test specified in MIL-STD-810, method 500.1, procedure I, except that:

a. First, the assembly in a non-operating condition, shall be exposed to an atmospheric pressure equivalent to 40000 feet above sea level for a period of not less than 1 hour.

b. Second, with the chamber pressure increased to an atmospheric equivalent to 10000 feet above sea level, subject the assembly to the performance tests of table VII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:29Z
Check the source to verify that this is the current version before use.

4.6.4.8 Dust. Subject the assembly to the dust test specified in MIL-STD-810, method 510.1, procedure I. At the conclusion of this test, subject the assembly to the performance tests of table VII.

4.6.4.9 Salt fog. Subject the assembly to the corrosion test specified in MIL-STD-810, method 509.1, procedure I. At the conclusion of this test, subject the assembly to the performance tests of table VII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:29Z
Check the source to verify that this is the current version before use.

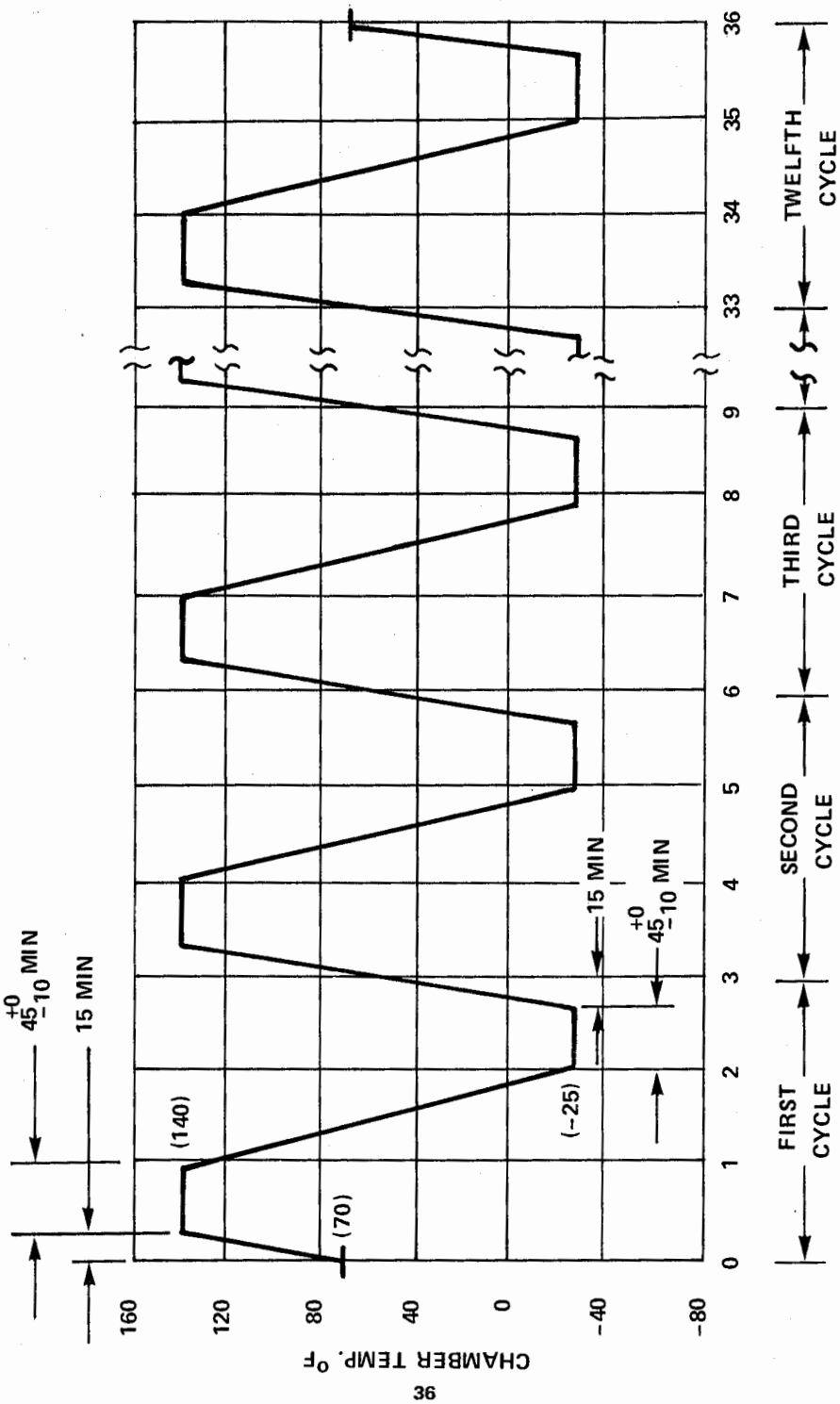


FIGURE 10. Temperature cycling.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:29Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Packaging, packing and marking. Packaging, packing and marking shall be in accordance with packaging data sheet P11743121. The level of protection shall be as specified in the procurement document. (See 4.5.6)

6. NOTES

6.1 Intended use. The Control, Commander's Sight and Receiver-Transmitter, Laser Ranging is intended to be used on the Laser Fire Control Rangefinder: AN/VVG-2.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Special tests and examination as required the type of inspection required, and the activity responsible for testing and its facility.

c. Applicable packaging Data Sheet number (see 5.1).

d. Selection of applicable levels of preservation, packaging and packing.

e. Applicable stock number.

f. First article inspection requirements.

6.3 Previous revisions of MIL-C-48490B was for the US Army Armament Research and Development Command's internal use only, and were not forwarded to the Naval Publications and Form Center for publication.

Custodian:
Army-AR

Preparing Activity:
Army-AR
Project No: 1240-A770

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:29Z
Check the source to verify that this is the current version before use.